Alejandro Baratas

AI Technical Lead | Agentic Engineering | Enterprise Modernization

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Profile

AI technical lead with 11 years in software engineering, from GIS and enterprise Java to agent-assisted delivery. I turn ambiguous work into source-grounded specifications, bounded implementations and reviewable evidence. At knowmad mood I contribute to an AI CoE and help teams apply project context, specialist agents and human approval to engineering work. I publish the method and its limits, including negative evaluation results.

Professional experience

knowmad mood | **Tech Lead AI** Mar 2026 – present

- Contribute to AI CoE architecture, adoption and hands-on engineering across agent workflows, delivery methods and governance. Build reusable patterns with colleagues, with human review as a release boundary.
- Helped shape a project-aware engineering nucleus for a cybersecurity-sector team: captured repository context, defined specialist roles, connected agents to inspectable PRs and adapted specification-driven development to the team workflow. Adoption scope is reported by the delivery team.
- Helped build and challenge Alejandría for traceable requirements, a source-linked modernization Factoría, Engineering Context for focused retrieval and KMM approval controls. These are separate company assets with bounded local checks, not one deployed platform.

Ayesa | **AI Technical Leader and AI Champion** Nov 2024 – Feb 2026

- Led AI integration and enablement work across engineering teams, combining architecture, pilots and hands-on delivery. Designed LLM integration, orchestration and governance approaches for enterprise use cases.

NTT DATA | **Technical Leader, Platform and AI Initiatives** Mar 2022 – Oct 2024

- Led architecture and technical decisions for enterprise platforms, cloud migration and microservices; mentored engineers and supported AI pilot work.

Hewlett Packard Enterprise | **Java Software Engineer** Feb 2019 – Feb 2022

- Built and operated Java backends and network-management systems in performance-sensitive enterprise and telecom environments.

Grupo AVALON | **Java Software Engineer** Apr 2017 – Jan 2019

- Evolved public-sector applications, including legacy Struts modernization, full-stack features and SQL-heavy workflows.

Repsol / Accenture | **GIS Software Engineer** Aug 2015 – Feb 2017

- Built geospatial tools and automated spatial workflows for industrial systems using ArcGIS, JavaScript, Python and FME.

Esri España | **GIS Systems Administrator** Apr – Jul 2015

- Supported ArcGIS platform administration and services for enterprise and public-sector projects.

Selected technical evidence

- SDD quality gates: helped develop an OpenSpec-based kit within the knowmad mood AI CoE. In two scored comparisons of plans on public code, its approach lost to an agent baseline (23–28 and 26–28 out of 40; AI judges, partial blinding). The negative result drove a clearer readiness and review process. A separate source-hook audit passed 225 simulated bash/dash cases.
- Banking modernization PoC: contributed a corpus-first, deterministic approach for a bounded reference module. An internal registry contained 23,811 evidence entries; 22,995 linked file, line and SHA-256. Two runs compared 4,338 files with no defects in the defined reproducibility check. This does not establish functional parity, client acceptance or financial savings.
- Public-sector preparation: helped harden a project-specific AI assistance package through 12 AI-led audit rounds and 36 documented corrections. July records show preparation and review, not later client deployment.

Technical focus

Agent workflows and review gates · AI governance and approvals · Source context and retrieval · Java and Spring · Python and FastAPI · TypeScript and Next.js · APIs and microservices · GIS and spatial data

Education

Degree in Geography and Territorial Planning | Universidad Complutense de Madrid | 2011–2015

ArcGIS technical and development master programme | Esri España | 2015–2016

Selected public work

Engineering method and case studies: alejandrobgn.vercel.app/engineering · alejandrobgn.vercel.app/work

Public agent-platform prototype and source: github.com/jandromani/A2A · github.com/jandromani/alejandrobgn